Fibre Optic Active Test Unit
SPS-8100
Ship Simulator
To Test Shore Systems

USER MANUAL

Issue	Detail	Date
1	Original	20/03/03

Q.1006.083

This page left intentionally blank

WARNING:

DO NOT REMOVE THE LID ON THE ACTIVE TEST UNIT IN HAZARDOUS AREAS.

This page left intentionally blank

CONTENTS

1. INTRODUCTION

2. HANDLING & CONNECTION

3. OPERATION

3.1 Pre-connection LED and Battery Test

3.2 ESD Testing

3.3 Telephone Testing

3.3.1 Channel 2 Hotphone
3.3.2 Channel 3 Telephone
3.3.3 Channel 4 Telephone

4. MAINTENANCE

4.1 Battery Charging

4.2 Battery Replacement

4.3 Handset Replacement

5. RECOMMENDED SPARES

5.1 Supplied Standard Spares

5.2 Optional Spares

6. SPECIFICATION

7. DRAWINGS LIST

Abbreviations used in this manual:

ATU	Active Test Unit
CCR	Cargo Control Room (or Central Control Room)
ESD	Emergency Shut Down
FO	Fibre Optic
PCB	Printed Circuit Board
SSL	Ship Shore Link

Warnings throughout this manual are highlighted in red.

1. INTRODUCTION

The Seatechnik SSL system comprises an Electric and a Fibre Optic, ESD and Telecommunications system. The testing of the FO system was originally tested by use of a Passive Loopback Connector, and was simply a brass mating connector that fitted into the system connector to be tested. Internally, there are FO patch leads connecting cores 1 to 2, 3 to 4 and 5 to 6.

Signal	Direction	Fibre Optic Core Number
Audio Telecommunications	Ship-shore	1
Audio Telecommunications	Shore-ship	2
ESD	Ship-shore	3
ESD	Shore-ship	4
Spare	Ship-shore	5
Spare	Shore-ship	6

When the Passive Loopback Connector is connected, the transmitted Audio signal from the shore on core 1 is fed back into core 2 and received by the shore. The TEST button on the Control & Alarm Module is used to test the return signal at the appropriate frequencies. The shore ESD signal on core 3 is fed back into core 4 and monitored by the shore, i.e., the shore monitors its own ESD 'out' signal.

The tests mentioned above are very limited and prove the Fibre Optic signal path only and not the full functionality of the system. The use of the Active Test Unit allows the end user to test not only the FO signal path, but also the audio and ESD signals on the shore fully, prior to connecting to the shipside system.

It is recommended that the shore side 'SSL' system be tested with the ATU, at least 24 hours prior to a ship berthing with the shore terminal.

2. HANDLING & CONNECTION

The component parts of this product are manufactured from Die Cast Aluminium, Nitrile Buna, ACETAL, Brass, Polycarbonate, Polyurethane and PVC. The performance of these materials shall be considered with regard to attack by aggressive substances that may be present in hazardous areas.

The ATU can be operated from inside the supplied aluminium box or can be removed from the box for easier handling.

Use the handle on the end on the ATU for carrying it around the ship. Take extreme care in handling the ATU. Do not drop the ATU as damage may occur to the enclosure and may create a safety hazard.

a) Ensure that the ATU is placed in suitable position close to the FO connector at the jetty, so that it can be operated properly, and, is not in a position such that it will fall and be damaged, or cause injury to any personnel.

b) Remove the caps from both FO connectors. Align connectors with each other using the keyways marked by red paint and mate the two using the plastic covered retaining ring on the ATU connector.

c) Operate ATU as described in section 3.

d) After use, disconnect the FO connectors and ensure that the caps are securely placed on to their relative connectors. This ensures that the FO cables ends are kept free of dust and dirt as best as possible.

3. OPERATION

The following tests in section 3 must be carried out with the enclosure lid in place. Only use the ATU in hazardous areas as approved by the ATEX certificate.

Note that during the tests carried out in sections 3.2 and 3.3, the common alarm from the SSL to CCR will be activated because of the loss of the telephone channel frequency carriers not being tested at that time, i.e., the ATU can only transmit one carrier at a time. This alarm should be ignored or overridden.

3.1 Pre-connection LED and Battery Test

a) Turn the 3-way ESD switch from the 'OFF' to the 'NORMAL' position to power up the ATU, and observe the status of the LEDs. If the battery has some charge in it, then some of the LEDs will illuminate depending upon the position of the other switches. If none of the LEDs illuminate, or briefly illuminate, then the battery may be totally discharged. If so, go to section 4.1.

b) If some of the LEDs are illuminated then check the status of the 'BATTERY LOW' LED. If this is illuminated then the battery requires charging – go to section 4.1. If this LED is 'OFF', then the charge level is above the charge level threshold, but there is no indication as to how much charge is stored in the battery, and hence the period of time that the ATU will function.

c) Turn the 5-way switch to the 'LED TEST' position and observe that all LEDs illuminate. If some LEDs do not illuminate then further investigation is required. It is likely to be the failure of a component or components on the LED board CB9113.

3.2 ESD Testing

Ensure that the FO connector is connected to the shore side connector, as described in section 2.

a) Turn the 3-way ESD switch from the 'OFF' position to the 'NORMAL' position. The ATU should now be transmitting the correct ESD signal to the shore SSL, and the CCR should be receiving a healthy ESD signal.

b) Turn the 3-way ESD switch to the 'ESD' position and observe that an ESD trip has occurred in the CCR.

c) Repeat tests in a) and b) again.

d) Ensure that the CCR is transmitting a healthy ESD to the ATU. Observe that the 'ESD NORMAL' LED is illuminated and the 'ESD' LED is off.

e) Generate an ESD trip from the CCR, and observe on the ATU that the 'ESD' LED is illuminated and the 'ESD NORMAL' LED is now off.

f) Repeat tests in d) and e) again.

g) Set the ESDs in the ATU and CCR to healthy. The 'ESD NORMAL' LEDs should now be illuminated. Now disconnect the FO connector from the shore side connector, and observe that the 'ESD' and 'ESD SIGNAL FAULT' LEDs in the ATU and the SSL are now illuminated.

This concludes the ESD tests.

3.3 Telephone Testing

Ensure that the FO connector is connected to the shore side connector, as described in section 2.

3.3.1 Channel 2 Hotphone

Ensure that the Hotphone mode is selected and not the Private Line for this test.

a) Turn the 'ESD' switch to the 'NORMAL' position, i.e., the ATU is powered up. Ensure that 'ON HOOK' is selected.

b) Select 'CH2' on the ATU, and observe that the 'RX CARRIER NORMAL' LED in the ATU is illuminated and the 'RX CARRIER ABNORMAL' LED is off. The channel 2 Ship to Shore 'NORMAL' LED should be illuminated and the other 3 channels should indicate ABNORMAL.

c) Move the switch to the 'Hotphone Call ' position. The speaker of Hotphone in the CCR should become active and allow voice communication. When answered, return the switch to the normal position and check speech path by talking to each other.

d) With the switch in the normal position, press the 'SIGNAL' button on the CCR Hotphone and observe that the 'SIGNAL CHANNEL 2' LED on the ATU illuminates. On observing this, lift the ATU handset, wait for the CCR to communicate, then check speech path.

3.3.2 Channel 3 Telephone

a) Select 'CH3' on the ATU.

b) Select 'CH3' on the ATU, and observe that the 'RX CARRIER NORMAL' LED in the ATU is illuminated and the 'RX CARRIER ABNORMAL' LED is off. On the system under test the channel 3 Ship to Shore 'NORMAL' LED should be illuminated and the other 3 channels should indicate ABNORMAL.

c) Move the switch to the 'OFF HOOK' position. Lift the ATU handset and note the presence of a dial tone from the PABX. Return the switch to normal position.

d) Arrange for the CCR to call the Channel 3 PABX extension and observe that the 'SIGNAL CHANNEL 3' LED on the ATU illuminates. On seeing this lift ATU handset, move the switch to the off hook position, establish communication with the CCR and check the speech path.

3.3.3 Channel 4 Telephone

a) Select 'CH4' on the ATU.

b) Select 'CH4' on the ATU, and observe that the 'RX CARRIER NORMAL' LED in the ATU is illuminated and the 'RX CARRIER ABNORMAL' LED is off. On the system under test, the channel 4 Ship to Shore 'NORMAL' LED should be illuminated and the other 3 channels should indicate ABNORMAL.

c) Move the switch to the 'OFF HOOK' position. Lift the ATU handset and note the presence of a dial tone from the PABX. Return the switch to normal position.

d) Arrange for the CCR to call the Channel 4 PABX extension and observe that the 'SIGNAL CHANNEL 4' LED on the ATU illuminates. On seeing this lift ATU handset, move the switch to the off hook position, establish communication with the CCR and check the speech path.

This concludes the telephone channel tests.

4. MAINTENANCE

The ATU has no user serviceable parts except for the battery and handset. If either part requires changing, then strictly follow the instructions below. Any deviation away from the instructions will invalidate the certification of the ATU.

It is recommended that the ATU be visually inspected before each use. Check for damage to the FO connector and cable, the handset and cable, and particularly to the enclosure. If there is any damage to the enclosure or if there is any doubt, then the ATU should not be used. The ATU must be sent back to Seatechnik or one of its agents for inspection and repair.

WARNING: Do not remove the lid on the ATU in a Hazardous area.

To remove the lid, back off the grub screw found on the top of the lid. Turn the lid anti clockwise until it can be removed. This will take approximately 8 full turns.

4.1 Battery Charging

WARNING: Only use the battery charger supplied with the ATU.

- The charger is for indoor use only.
- The charger must not be covered up as it may overheat.
- There are no user serviceable parts inside.
- Do not open the cover, as high voltages exist within the charger.

a) Ensure that the ATU is switched 'off'. Remove the lid.

b) Connect the jack plug on the charger flying lead into the jack socket on the left-hand side of the LED legend plate. Then plug the charger into the mains socket and turn on the mains supply. Do not connect the charger before the jack plug is fitted to the ATU.

c) Observe the tri-colour LED on the charger. The LED indicates the status of the charging process. Orange indicates discharged state and the charge current is at maximum. Yellow indicates charge voltage level is at 95% of final voltage level; charge current is less than maximum. A 4-hour timer is also started and will stop charging when it times out or the maximum charge level is reached. Green is fully charged and the charging has stopped.

d) Once fully charged, as indicated by the LED on the charger turning green, turn off the charger first and then remove the charger jack plug from the ATU. Replace the lid as instructed below.

The charging process takes approximately 3 hours on a new fully discharged battery. Note that the charging time may increase with time and usage.

The number of charge – discharge cycles is approximately 500.

If the amber or green LEDs are never illuminated, then the battery may need changing, or the charger may be faulty.

4.2 Battery Replacement

WARNING: only replace the battery with the same type – SAFT Part No. MP 144350.

Refer to drawings SHS-8110-153 and SHS-8110-402 during this procedure.

a) Ensure that the ATU is switched 'off'. Remove the lid.

b) Turn the brass 'PCB Assembly Spindle' anti-clockwise until the PCB assembly is freed. Then, very carefully, remove the whole PCB assembly from the enclosure, without placing any stress on the cables, and place it on a suitable flat work area. Ensure that the FO cores and ribbon cables are not damaged during this process. Any damage to the cables may cause the ATU to inoperable.

c) Carefully disconnect the 4 FO connectors on cores 1 to 4 from the FO&ESD Interface Board CB9111, then remove the remaining two spare cores 5&6 from the FO mounting bracket. Note the FO core numbers in relation to the positions of the FO connectors on the PCB. The PCB Assembly can now be moved to a workbench area

d) Disconnect the 2-way MOLEX connector (2 core cable - red and black- connected to the battery box) from CON4 on PSU board - CB9114.

e) Disconnect the 3-way MOLEX connector (3 core ribbon cable connected to the Zener Board) from CON3 on the PSU board - CB9114.

f) Now disconnect the 12-way MOLEX connector (11 core ribbon cable connected to the 3 switches) on CON1 on the PSU board - CB9114. The PCB Assembly can now be moved to a workbench area.

g) Carefully remove the circlip from the 'PCB Assembly Spindle' and then pull the spindle out of the assembly. The boards can now be released.

h) Carefully remove the LED legend plate and then each PCB starting from the top, by unscrewing the 4 retaining screws and then placing the PCB to one side. Note the position of each board for reverse replacement later on.

i) Remove the lid from the battery box. Now remove the battery from the box. The crimped wires on the MOLEX connector will have to be removed from the connector in order to feed the cable through the grommet.

j) Fit a new battery of the specified type. Feed the wires through the grommet and insert wire crimps into correct position on the MOLEX connector.

k) Reassemble the PCBs in their correct position, ensuring that each PCB is fixed into position by the 4 fixing screws. Drawing SHS-8110-151 shows the PCB assembly order.

l) Fit the 'PCB Assembly Spindle' by placing it through the holes of the PCBs. Now fit the circlip on the underside. This is item 46 as shown on drawing SHS-8110-151.

m) Reconnect the 3-way MOLEX connector to CON3 on the PSU board – CB9114.

n) Reconnect the 12-way MOLEX connector to CON1 on the PSU board – CB9114.

o) Reconnect the 2-way MOLEX connector from the battery to CON4 on the PSU board – CB9114.

p) Move the assembly closely to the enclosure and carefully reconnect the FO cores 1 to 4 into their correct sockets on the FO ESD Interface board CB91111. Then connect the spare cores 5 & 6 on to the FO bracket fixed to the PCB Base Plate.

q) Now position the PCB Assembly into the enclosure taking extreme care not to damage any of the cables or FO cores, and observing the correct routing and fixing of the ribbon cables into their cable clamps. See drawing SHS8110-153.

r) Engage the locating pins on the underside of the 'PCB Stack Base Plate' into the alignment holes in the 'PCB Mounting Plate' and then rotate the brass 'PCB Assembly Spindle' clockwise until fully home.

s) The enclosure lid can now be placed into position and tightened as described at the end of section 4.

4.3 Handset Replacement

WARNING: only replace the Handset with the same type – GAITRONICS-Part No. 100-02-0032-002. Only clean handset with damp cloth.

Refer to drawings SHS-8110-153 and SHS-8110-401during this procedure.

a) Ensure that the ATU is switched 'off'. Remove the lid.

b) Turn the brass 'PCB Assembly Spindle' anti-clockwise until the PCB assembly is freed. Then, very carefully, remove the whole PCB assembly from the enclosure, without placing any stress on the cables, and place it on a suitable flat work area. Ensure that the FO cores and ribbon cables are not damaged during this process. Any damage to the cables may cause the ATU to inoperable.

d) Carefully disconnect the 4 FO connectors on cores 1 to 4 from the FO&ESD Interface Board CB9111, then remove the remaining two spare cores 5&6 from the FO mounting bracket. Note the FO core numbers in relation to the positions of the FO connectors on the PCB. The PCB Assembly can now be moved to a workbench area

e) Carefully remove the 4 wires connected to CON 2 on the Zener Board – CB9115. Loosen P-clip and remove cable. Back off the cable gland and remove the telephone cable from the enclosure.

f) Insert the cable of the new telephone through the cable gland and P-clip, and then reconnect the 4 cores to CON2 on the Zener Board – CB9115. Once connected tighten the cable gland appropriately.

g) Carefully reconnect the FO cores 1 to 4 into their correct sockets on the FO ESD Interface board CB91111, and connect the spare cores 5 & 6 on to the FO bracket fixed to the PCB Stack Base Plate.

h) Now position the PCB Assembly into the enclosure taking extreme care, not to damage any of the cables or FO cores, and observing the correct routing and fixing of the ribbon cables into their cable clamps. See drawing SHS8110-153.

i) Engage the locating pins on the underside of the 'PCB Stack Base Plate' into the alignment holes in the 'PCB Mounting Plate' and then rotate the brass 'PCB Assembly Spindle' clockwise until fully home.

j) The enclosure lid can now be placed into position and tightened as described at the end of section 4.

Replacing the lid is achieved by placing it in position then turning it clockwise, until the lid face is hard up against the body face. It is imperative that the grub screw is then fastened in order to prevent the lid backing off and accidentally opening when in use.

5. RECOMMENDED SPARES

5.1 Supplied Standard Spares

Qty	Item	Part No.
2	Circlip 'E' type 5mm	289-483
1	15mm Spacer	606-692
2	18mm Spacer	221-156
5	0.1" KK Crimps	467-598
2	M3 x 10 pan head screw	189-311

5.2 Optional Spares

Qty	Item	Part No.
1	SAFT Battery	MP144350
1	Handset	100-02-0032-002
1	Clamp	143-911

6. SPECIFICATIONS

Enclosure

- Size
  Height – 200mm
  Length – 260mm
  Width – 212mm
- Weight 14Kg
- Certification SIRA 03ATEX1080
- Compliance Eex d ib IIB T6
  Ex II2 G
  CE 0518

Environmental Properties

- Temperature range 5°to 50° non-condensing
- IP Rating IP66

Power Supply

- Input Voltage to ATU 8.6Vdc for ATU Boards
- Power Rating <4W
- Running Time Approximately 1.5 hours before 'Battery Low LED' illuminates. Reserve Time approximately 45 minutes
- Charge Time Approximately 3 hours

Battery

- Type Rechargeable Lithium Ion
- Manufacturer SAFT
- Part No. MP144350
- Nominal voltage 7.2V
- Average Capacity 2300 mAh
- Protection Internal protection against over-charge, over-discharge and over-current

Battery Charger

- Type 9461
- Manufacturer Mascot Electronics A/S
- Part No. 9461 0000 67
- Input Voltage 230VAC, 50-60Hz, 0.5A
- Output Voltage 8.2VDC (for charging 7.2V nominal Li-Ion battery) @ 2.3A
- Charging Temp Range 0 to +40C

Handset

- Manufacturer Gai-Tronics
- Part No. 100-02-0032-002
- Earpiece Besson M4-AW-Z150
- Microphone Besson M4-AW-Z150

7. DRAWINGS LIST

Drawing Number	Description
SPS-8100-100	ATU Ship Simulator
SHS-8110-151	PCB Stack General Assembly
SPS-8100-153	PCB Layout and Front & Rear Panel Details
SHS-8110-180	Hazardous Warning Label-1
SHS-8110-183	Warning label-2
SHS-8110-401	Zener Barrier Enclosure Internal Wiring
SHS-8110-402	PCB Stack Battery Wiring

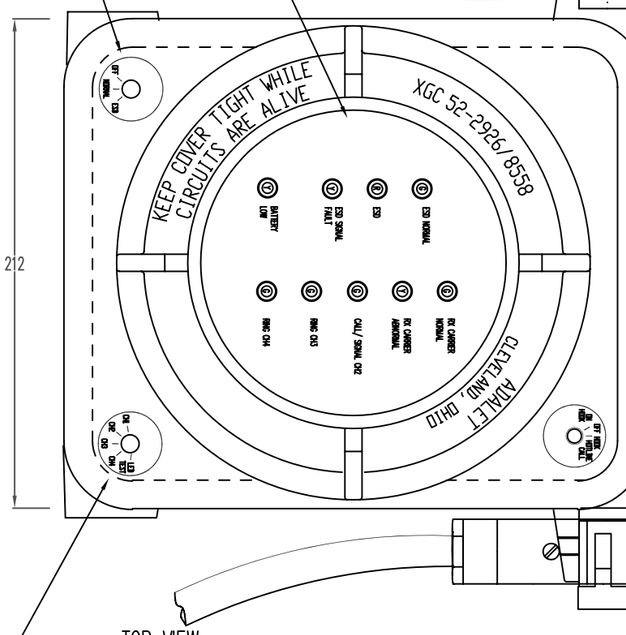
ADALET MINIATURE Ex'd' 3 POS SWITCH

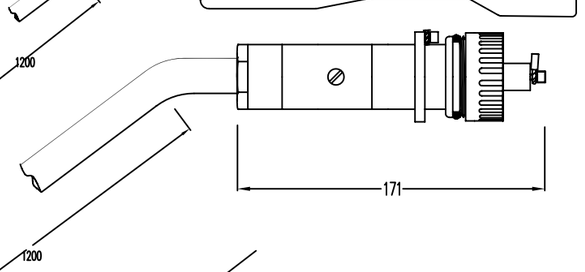
STOWED POSITION OF HANDSET

25MM TUBE CARRYING HANDLE

EXPLOSIONPROOF WINDOW
VIEWING OD 132mm

TELEPHONE HANDSET



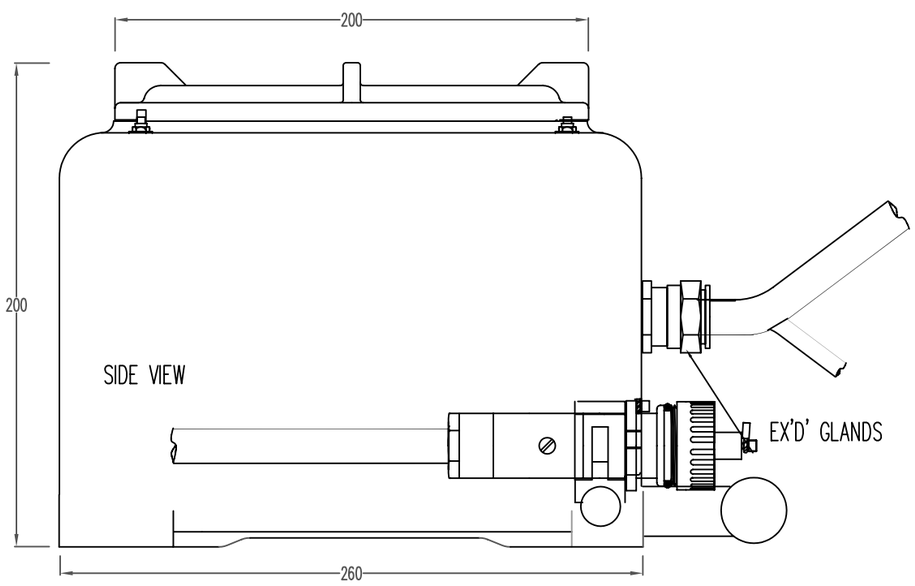


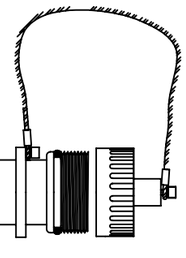
NFI SHIP-TYPE FURUKAWA 6-WAY COMPATIBLE FIBREOPTIC CONNR

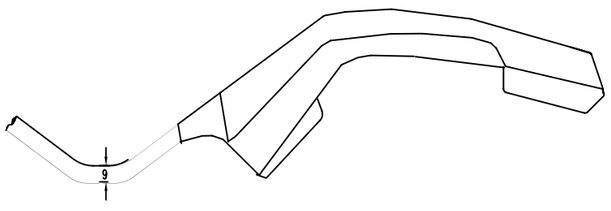
STOWED POSITION OF FO CONNECTOR

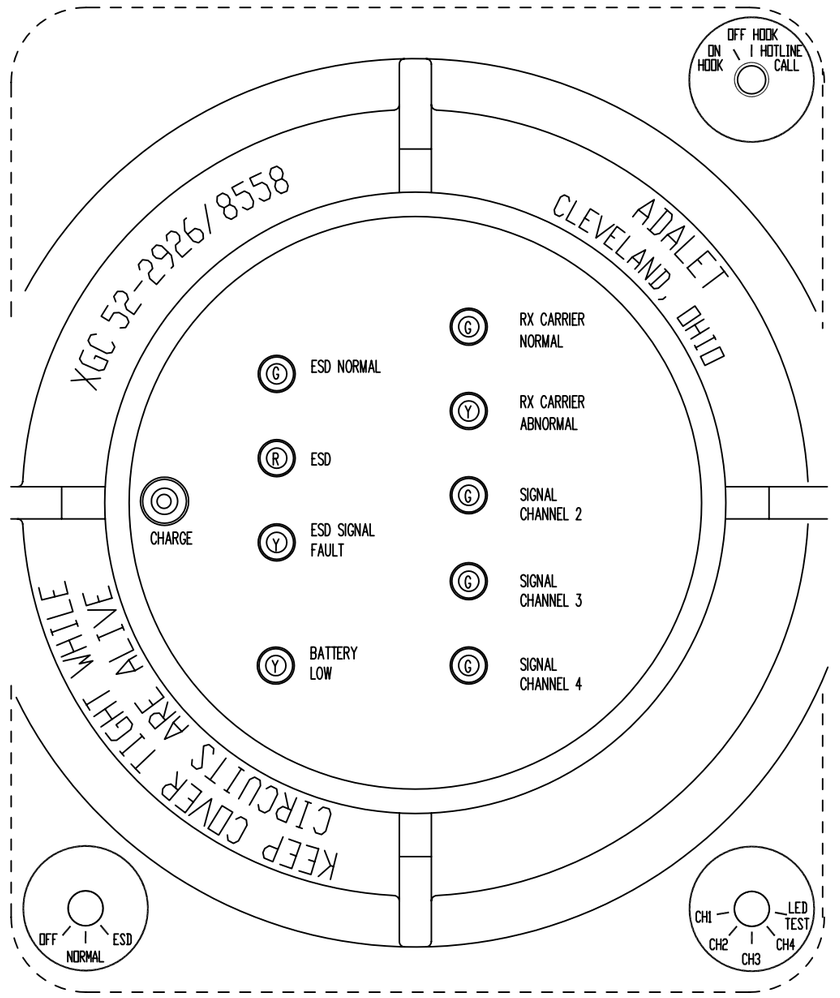
TOP VIEW

ADALET MINIATURE 5 POS SWITCH









LAYOUT OF CONTROLS AND INDICATOR LEDS

NOTES
UNIT INSTALLED IN ADALET CENELEC EX'D' EXPLOSIONPROOF XJAGC ENCLOSURE
WEIGHT APPROX 15KG

BATTERY LIFE APPROX 1.5 HOUR CONTINUOUS USE
SEE MANUAL FOR RECOMMENDED TEST PROCEDURE
HANDSET EX'IB'
CERTIFICATE No SIRA 03ATEX1080
EEx d ib IIB T6

ISSUE	DESCRIPTION	APPD.	DATE
2	REVISED CARRYING HANDLE/SWITCHING	RGC	04/12/02

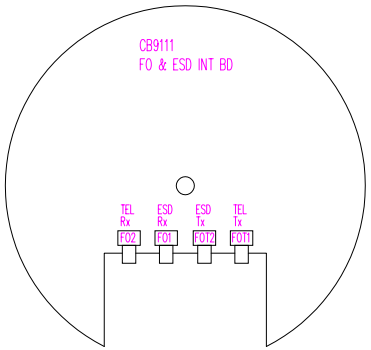
SeaTechnik
Deeside UK.
tel +44 (0) 1244 535787

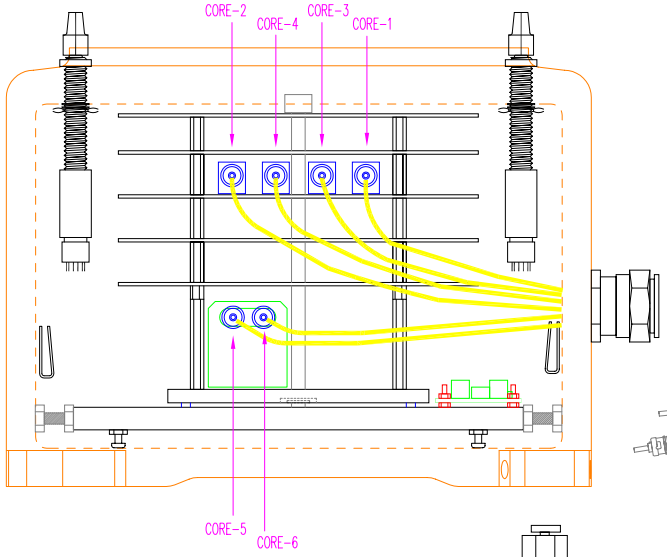
FIBREOPTIC SHIP-SHORE LINK
ACTIVE TEST UNIT
SHIP SIMULATOR (TEST SHORE)

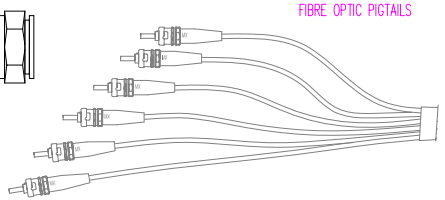
DRAWN	TRACED	CHECKED	APPROVED	DATE
RGC				24/03/01

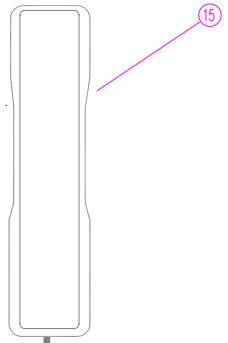
DRAWING No. SPS-8100-100-0-2

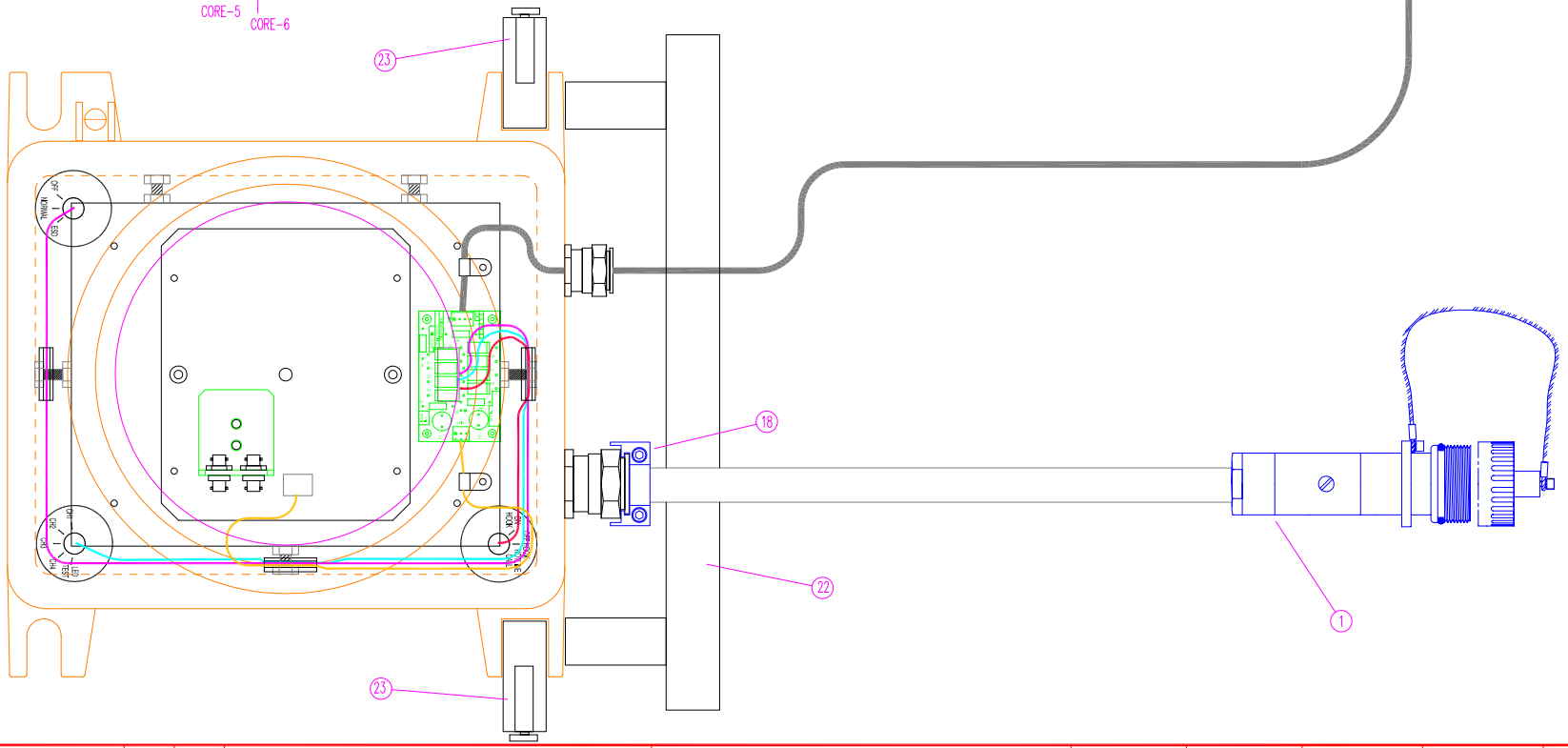
REV. 2











NOTES:
1 – THIS IS A 'SIRA' CERTIFIED PRODUCT. NO CHANGES PERMITTED WITHOUT REFERENCE TO SIRA.
2 – SEE PARTS LIST PL8100 FOR ITEM No PARTS
3 – THIS ASSY DWG USED FOR SPS8100 ONLY
4 – SEE DWG SHS-8110-400-0-1 FOR SWITCH WIRING DETAILS
5 – SEE DWG SHS-8110-401-0-1 FOR ZENER BARRIER CIRCUIT WIRING DETAILS

ISSUE	DESCRIPTION	APPD.	DATE

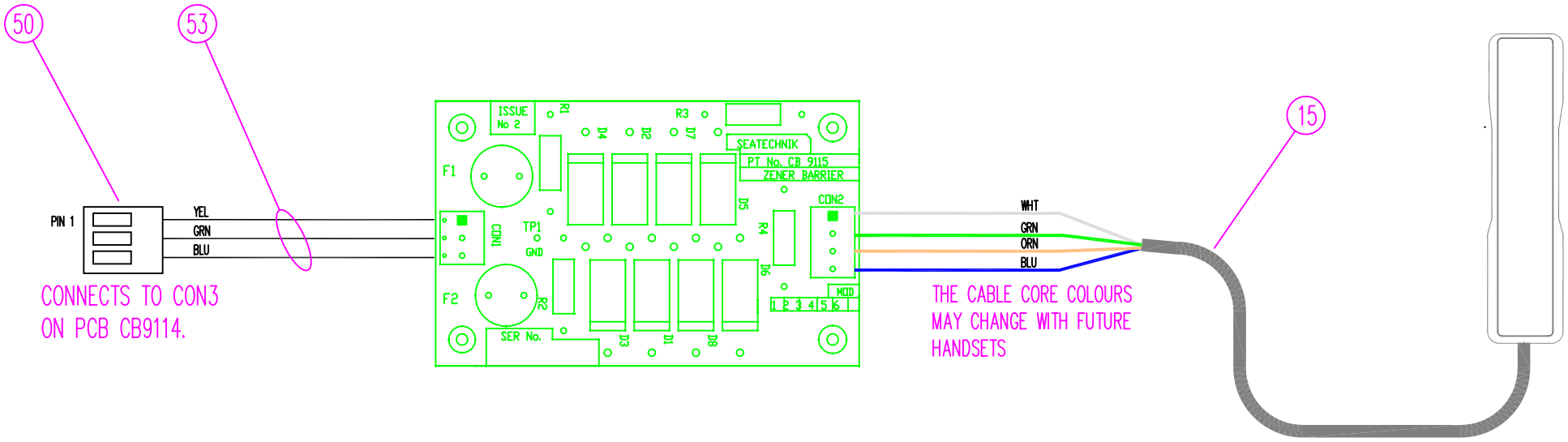
NFI SeaTechnik
Camberley & Deeside UK.
tel +44 (0) 1244 535787
+44 (0)1276 600200

TITLE
FO ACTIVE TEST UNIT
EXTERNAL COMPONENT ASSEMBLY
SHIP SIMULATOR

DRAWN		CHECKED	APPROVED	DATE
EJM		RGC	IMT	12/03/03

DRAWING No. SPS8100-153-0-1

REV. 1



NOTES:

1 – THIS IS A 'SIRA' CERTIFIED PRODUCT. NO CHANGES PERMITTED WITHOUT REFERENCE TO SIRA.

2 – SEE PARTS LIST PL8110 FOR ITEM No PARTS

3 – THIS ASSY DWG USED FOR SHS8110 & SPS8100 ATU

4 – SEE DWG SHS-8110-152-0-1 FOR WIRING ASSEMBLY DETAILS

5 – THE PCB MUST HAVE TWO LAYERS OF CONFORMAL COATING.

HANDSET SPECIFICATIONS

IMPEDANCE 150R
DC RESISTANCE 21R
INDUCTANCE 21mH

SEE DOCUMENT 'ATU HANDSET SPECIFICATIONS' FOR FURTHER DETAILS

ISSUE	DESCRIPTION	APPD.	DATE

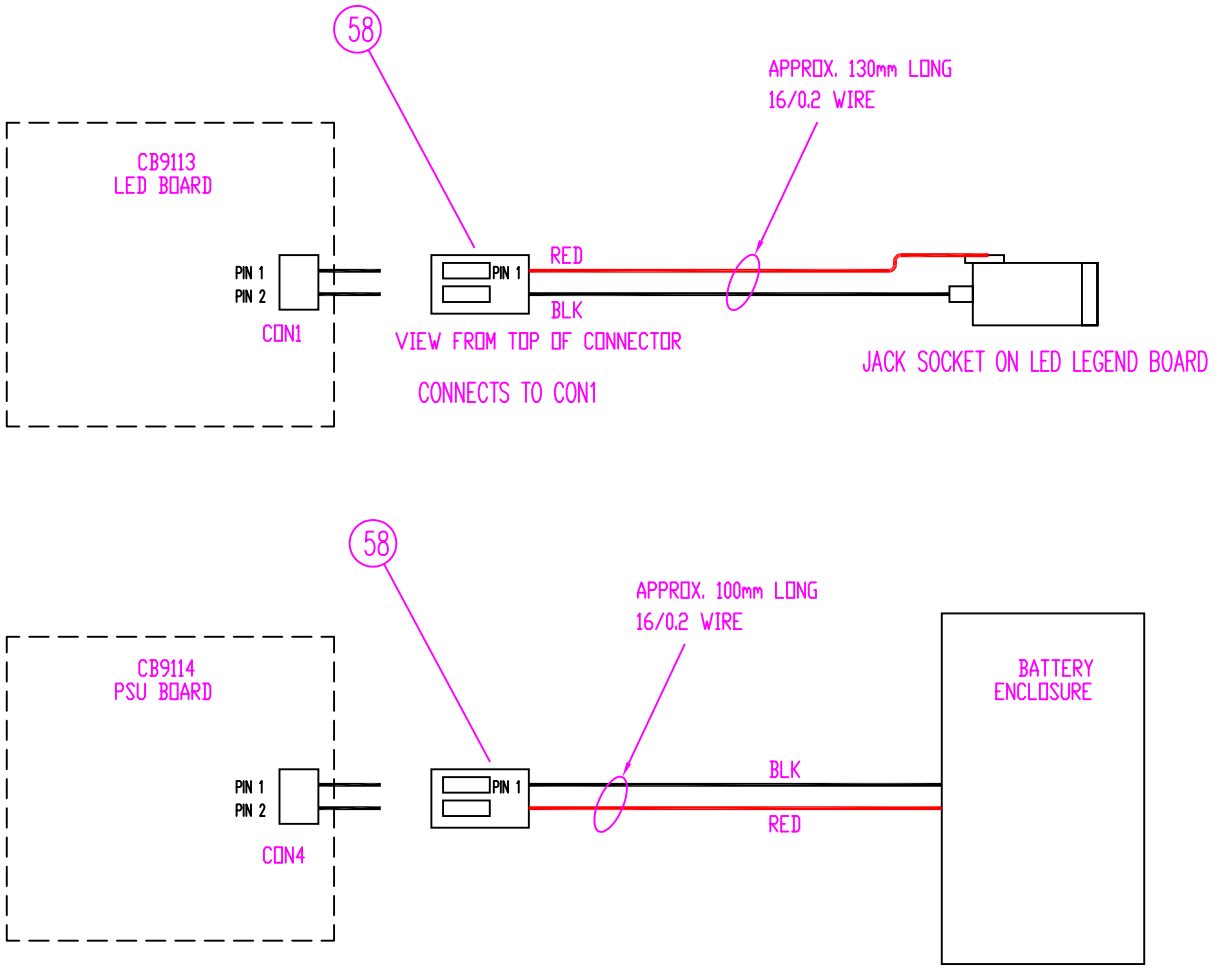
SeaTechnik
NFI
Camberley & Deeside UK.
tel +44 (0) 1244 535787
+44 (0)1276 600200

TITLE
FO ACTIVE TEST UNIT
ZENER BARRIER CIRCUIT
ENCLOSURE INTERNAL WIRING

DRAWN	TRACED	CHECKED	APPROVED	DATE
EJM		PSP	IMT	18/03/03

DRAWING No. SHS-8110-401-0-1

REV. 1



NOTES:

1 – THIS IS A 'SIRA' CERTIFIED PRODUCT. NO CHANGES PERMITTED WITHOUT REFERENCE TO SIRA.

2 – SEE PARTS LIST PL8110 FOR ITEM No PARTS

3 – THIS ASSY DWG USED FOR SHS8110 & SPS8100 ATU

ISSUE	DESCRIPTION	APPD.	DATE

SeaTechnik
NFI
Camberley & Deeside UK.
tel +44 (0) 1244 535787
+44 (0)1276 600200

TITLE

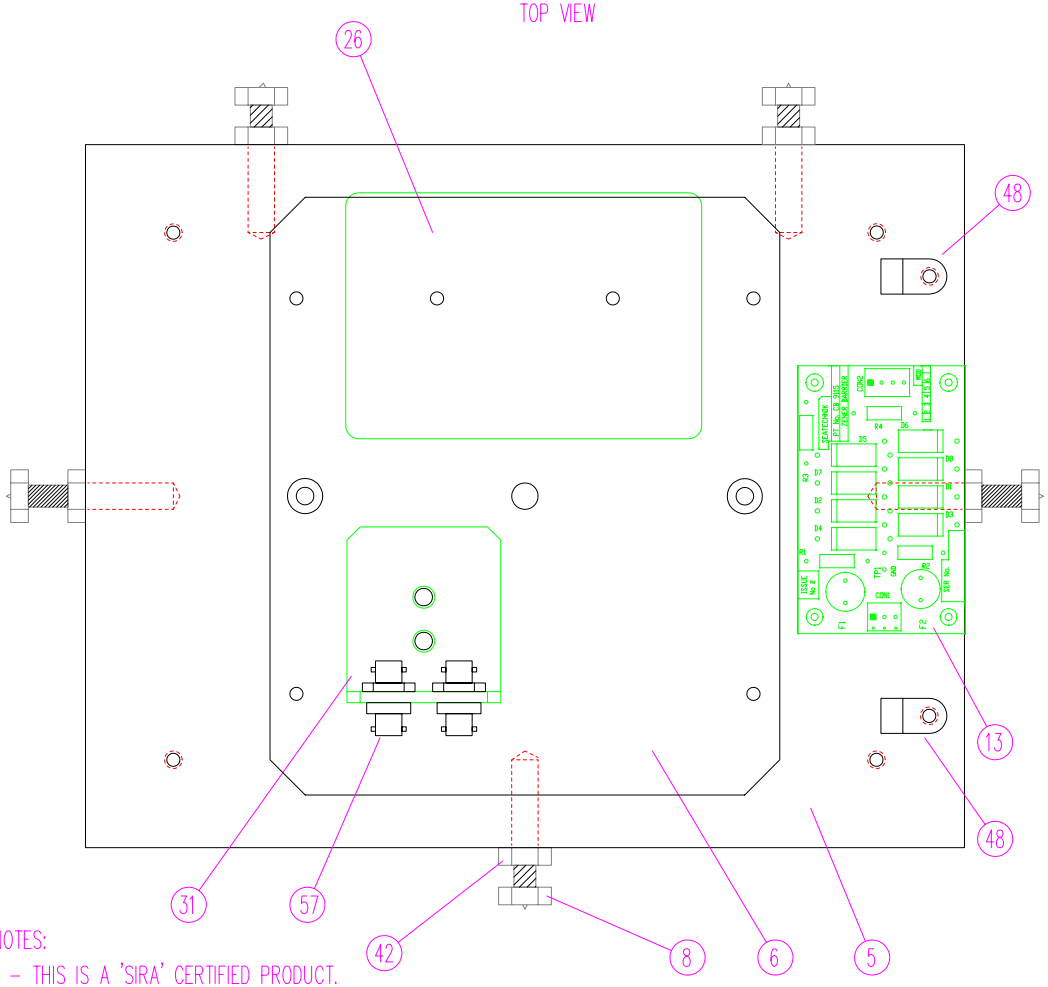
FO ACTIVE TEST UNIT
PCB STACK – BATTERY WIRING

DRAWN	TRACED	CHECKED	APPROVED	DATE
EJM		PSP	IMT	13/03/03

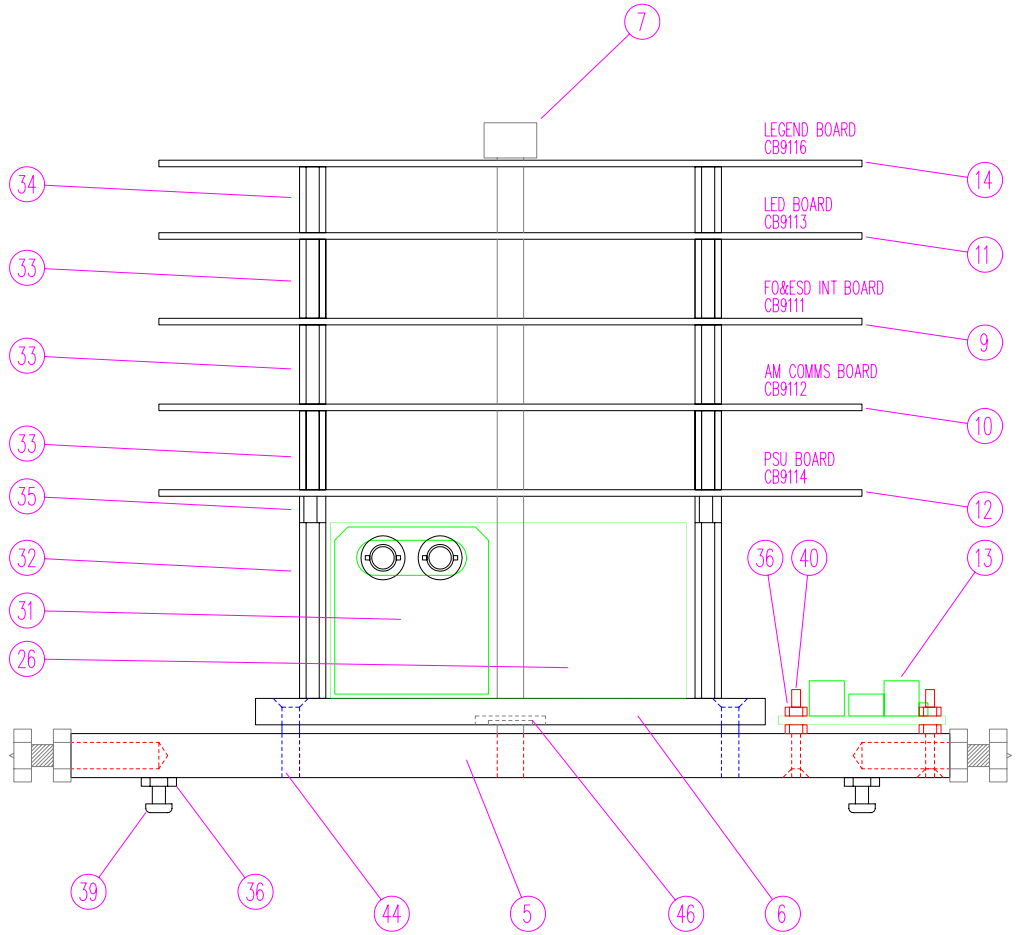
DRAWING No. SHS-8110-402-0-1

REV. 1

TOP VIEW



SIDE VIEW



NOTES:

1 – THIS IS A 'SIRA' CERTIFIED PRODUCT. NO CHANGES PERMITTED WITHOUT REFERENCE TO SIRA.

2 – SEE PARTS LIST PL8110 FOR ITEM No PARTS

3 – THIS ASSY DWG USED FOR SHS8110 & SPS8100 ATU

4 – SEE PROJECT BUILD & PARTS LIST FOR PCB VARIATIONS AND ISSUES.

ISSUE	DESCRIPTION	APPD.	DATE

NFI SeaTechnik
Camberley & Deeside UK.
tel +44 (0) 1244 535787
+44 (0)1276 600200

TITLE
FO ACTIVE TEST UNIT
PCB STACK – GENERAL ASSEMBLY

DRAWN		CHECKED	APPROVED	DATE
EJM		RGC	IMT	11/03/03

DRAWING No. SHS8110-151-0-1

REV. 1